SQL quick reference

Database Systems, VertexLearn. Core statements for SQL Server (T-SQL).

CRUD statements

`INSERT INTO t (a, b) VALUES (@a, @b)`	Create a new row. Use parameters, never string concatenation.
`SELECT a, b FROM t WHERE a = @a`	Read rows that match a condition.
`UPDATE t SET b = @b WHERE a = @a`	Modify existing rows. Always include WHERE.
`DELETE FROM t WHERE a = @a`	Remove rows. Always include WHERE.

Filtering, sorting, grouping

`WHERE col LIKE '%' + @s + '%'`	Pattern search.
`ORDER BY col DESC`	Sort results.
`SELECT TOP (5) ...`	Limit the number of rows (SQL Server).
`GROUP BY col HAVING COUNT(*) > 1`	Aggregate rows and filter the groups.
`COUNT, SUM, AVG, MIN, MAX`	Aggregate functions.

Joins

`INNER JOIN`	Only rows with a match in both tables.
`LEFT JOIN`	All rows from the left table; NULLs where the right has no match.
`t1 JOIN t2 ON t2.Id = t1.T2Id`	Join condition on the foreign key.

Keys and constraints

`PRIMARY KEY`	Uniquely identifies each row; cannot be NULL.
`FOREIGN KEY ... REFERENCES`	Links to a primary key in another table.
`ON DELETE CASCADE`	Delete child rows automatically when the parent is deleted.
`UNIQUE`	No duplicate values allowed (e.g. Email).
`CHECK (Score >= 0)`	Rule every row must satisfy.
`IDENTITY(1,1)`	Auto-incrementing number for new rows.
`SCOPE_IDENTITY()`	The IDENTITY value just inserted in this scope.